Simulation of the Motion of Astronomical Bodies

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Here's the outline for the report regarding the layout and process.

- Structured work:
  - Structured(1) - Stationary Sun and moving Earth
  - Structured(2) - Track period of Earth's orbit
  - Structured(3) - Track period of new planet based off student id number
- Structured work:
  - Allow Sun to move due to pull of Earth
  - Extend simulation to include all planets of the Solar System
  - Allow planets of Solar System to move clockwise and counter-clockwise to view how stability is affected
  - Add asteroids to simulation to examine behaviour of Kirkwood gaps

1 Abstract

"An abstract that summarizes your project and the key results/conclusions"

Using python, a simulation of the bodies of the Solar System was created

2 Usage of AI

3 Introduction

The motion of celestial bodies in the Solar System is primarily determined by the force of gravitational attraction between all the planets. This report aims to

It was assumed that all motion will take place on the x-y plane

4 Methods

The first script written aimed to simulate the motion of the Earth in orbit around the Sun. Since the mass of the Sun is much larger than that of the Earth ($\frac{M_{Sun}}{M_{Earth}} \approx 3.3 \times 10^5$), it was assumed that the affect of Earth's pull on the Sun was negligible and the Sun would remain at rest during the simulation.

The Sun was fixed at (0,0) and the Earth was placed 1 AU away from the Sun at a random angle to the horizontal axis. The Runge-Kutta method was used to numerically calculate the updated position and velocity of the Earth after each timestep. The simulation was run for 3 years at a timestep of 1 hour and the position of the Earth was plotted using matplotlib to give Figure 1.

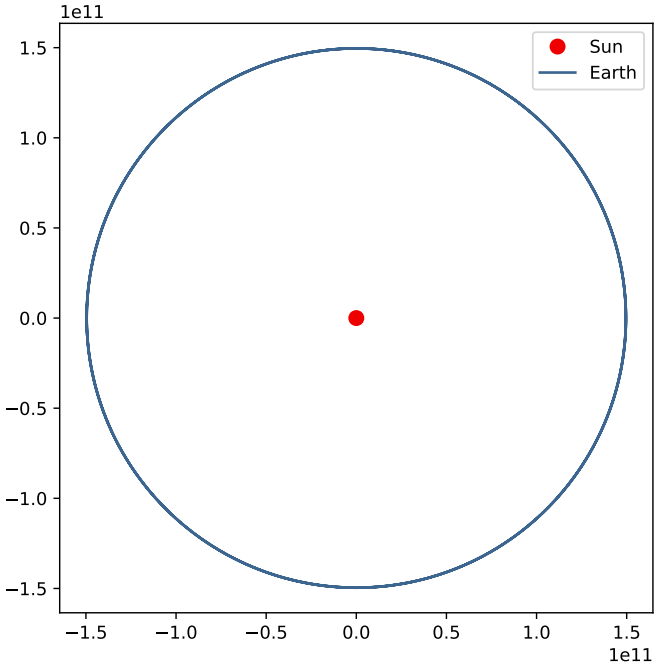


Figure 1: Structured 1 plot

Visually this behaviour seems accurate as we see the Earth travelling in a steady circular orbit. To measure the accuracy of the simulation it was decided that the period of Earth's orbit should be measured. This was achieved by measuring time between successive crossings of the positive X-axis. To determine whether an axis crossing had occurred, the programme used the following check:

```
1     if y_coord_prev * y_coord_now < 0 and x_coord_now >
          0:
```

where `y_coord_now` was the most recent y coordinate of the Earth and `y_coord_prev` was the y coordinate immediately before this.

Usage of Runge Kutta method

Sun and Earth model with stationary Sun

Orbital time tracker

Sun and Earth model with Sun able to move

Sun and arbitrary body model where initial velocity is given to body but not Sun.

Give Sun initial momentum to give system 0 total momentum

Add rest of bodies from Solar System into simulation

Examine Kirkwood gaps by only considering Sun, Jupiter and asteroids at various starting orbits

5 Results

"Your results"

"Discussion of results and conclusions"

6 Conclusion

Inaccuracies in this simulation may originate from a few sources, such as

- Circular orbits vs. elliptical orbits
- No orbital inclination
- Choice of simulation timestep

7 Appendix

"An appendix with a printout of your Python program(s)"